Multiplying Fractions Informally with Reasoning and Manipulatives

Mr. Dobson

Warm-up!

*Before bed I had ⅔ **of** my book left. This morning I read ½ **of** what was left.*

How much of the **whole** book I have I read?

Debrief

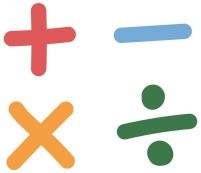
- ➔ In the question what are some keywords that can help us understand the problem?
- ➔ Is the amount read this morning more or less than half of the whole book?

Learning Goal

I can…

➔ Represent the multiplication of two proper fractions that are between 0 and 1

Success Criteria



I will…

- ➔ Be able to represent both fractions on the same whole using manipulatives
- ➔ be able to explain why the result of two fractions being multiplied is smaller
- ➔ be able to use mathematical terms like **whole, part, of** to explain my thinking.

Example: $\frac{2}{3} \times \frac{1}{4}$

Using the manipulatives with a partner see if we can apply our method for the book to this problem now.

https://toytheater.com/fraction-strips/

Your turn!

You and your partner will be given a task card.

1. You and your partner will try using the **manipulatives** to apply a method to try and solve the problem
2. Write a one sentence explanation such as:

> “I showed _ **of** a **whole,** then found_ **of** that **part**. The final result is _**of** the whole”

Share!

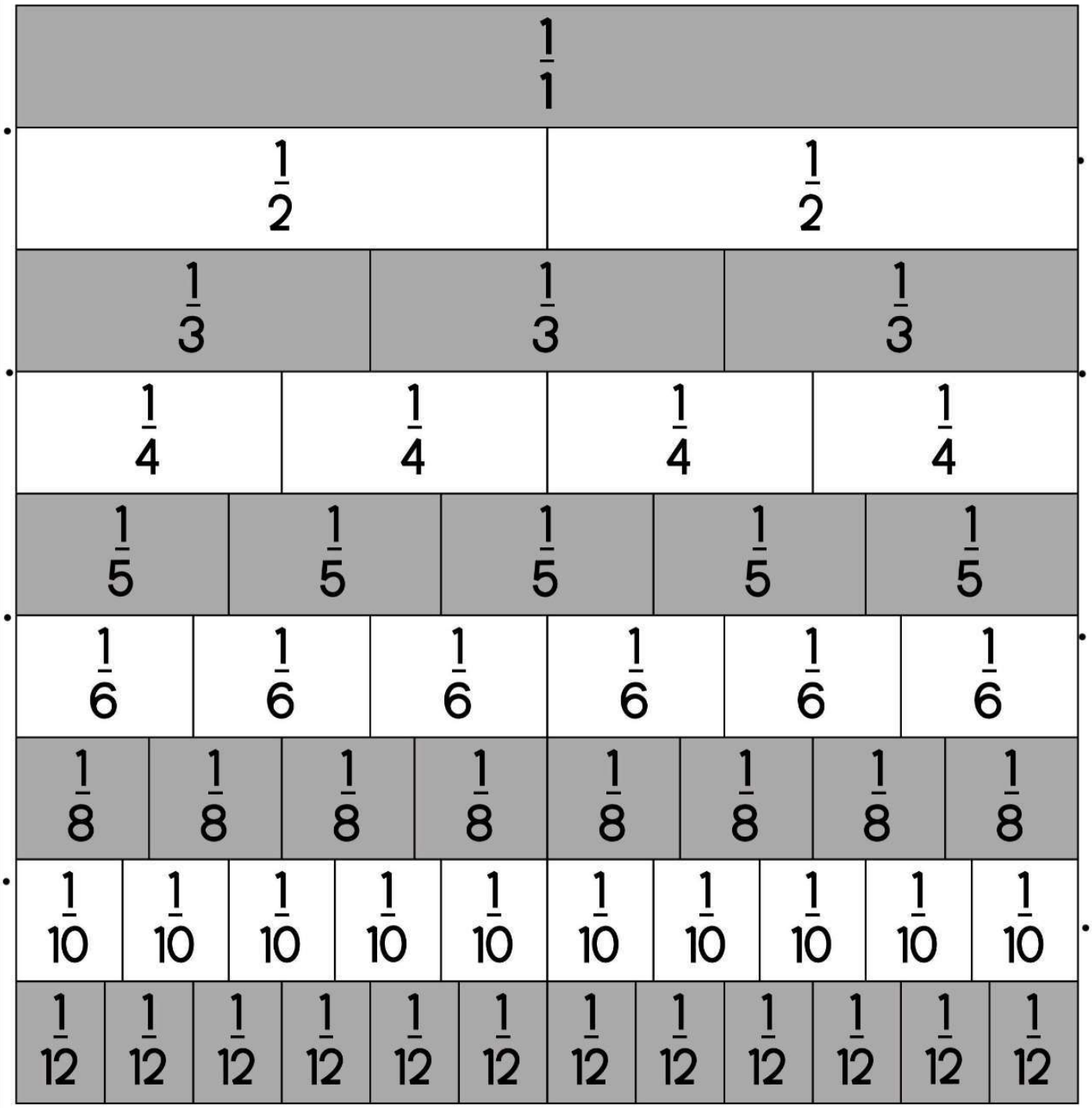
- ➔ What patterns did we notice between the methods?
- ➔ What happens when both fractions are less than 1?
- ➔ Does swapping the order change our answer?

Sticky Note Activity

Each student will have two sticky notes. On each sticky note answer the following,

1. When we multiply fractions using our manipulatives, what steps did we follow today?
2. What are you still wondering about multiplying fractions?

Have a good recess!


1/1
1/2 1/2
1/3 1/3 1/3
1/4 1/4 1/4 1/4
1/5 1/5 1/5 1/5 1/5
1/6 1/6 1/6 1/6 1/6 1/6
1/8 1/8 1/8 1/8 1/8 1/8 1/8 1/8
1/10 1/10 1/10 1/10 1/10 1/10 1/10 1/10 1/10 1/10
1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12
ourlittlesliceofheaven.com